

THE STRUCTURE OF MATHEMATICAL REALITY

Phase V — Chapter 16

Category Theory: The Mathematics of Mathematics

Note Template

Self-Study Curriculum

How to use this template

Complete these notes **after** the reading and the lecture. The goal is not to copy phrases from the text — it is to reconstruct the ideas in your own language. Leave every item blank on first pass; fill it in from memory, then check.

1. A category consists of

- a.
- b.
- c.
- d.

2. The key shift: instead of studying structures directly,

study _____.

3. Four examples of categories (from the lecture)

- a. Objects: _____ Morphisms: _____
- b.
- c.
- d.

4. A functor is

What it preserves:

Why it formalizes “bridge between fields”:

5. A natural transformation is

your best understanding: it's okay if this is rough

6. A universal property defines an object by

contrast with “defines an object by what it is”

7. The hierarchy

Objects $\rightarrow$ _____ $\rightarrow$ _____ $\rightarrow$ _____

Each level studies the _____ of the previous level.

8. A pattern I noticed across multiple chapters that I now

suspect has a categorical explanation:

describe it
